VIRELEO INTERN ENGINEERING REPORT • PROJECT 01

AqarFlow — Day 3 Daily Progress Report

Real Estate Management & Property Operations Platform • 7-Day MVP Execution

STATUS: PRODUCTION BUILD PASSING (37/37 ROUTES) LIVE DEPLOYMENT: VERCEL SYNCHRONIZED

SECURITY: STRICT RBAC & ISOLATION COMPLETE

Project Name	**AqarFlow (Project 01)**	**Submission Date**	**November 11, 2026 (Day 3 of 7)**
Parent Organization	**VIRELEO** (https://www.vireleo.me/)	**Development Model**	Individual Full-Stack Ownership (1 Intern Developer)
Current Version	**Version 1.3 — Enterprise RBAC, Architectural Polish & Minimalist Editorial UX**	**Primary Market Focus**	Pakistan Domestic (Islamabad, Lahore, Karachi, Rawalpindi) + Overseas Diaspora (UK, US, Canada)
Tech Architecture	Next.js 16 (App Router), React 19, Tailwind CSS v3.4, TypeScript, In-Memory DB + Multi-Persona RBAC	**Verification Score**	**100% Test Pass Rate (37/37 Routes Prerendered, 0 Hydration Errors, 0 Security Leaks)**
Live Deployment URL	https://aqar-flow-topaz.vercel.app	**GitHub Repository**	https://github.com/DiwakerPandey21/AqarFlow.git

1. Executive Summary & Day 3 Milestones Delivered

During **Day 3 of the 7-day MVP delivery cycle**, engineering focused on resolving critical user experience, visual branding, route-security, and role-based access control (RBAC) requirements identified during executive testing. AqarFlow has evolved into a zero-compromise, institutional-grade luxury real estate ERP and marketing platform. Key deliverables completed ahead of schedule include:

- **Hydration Error Eradication:** Resolved React hydration mismatch in `PropertyCard.tsx` caused by SSR server locale formatting differences on USD conversions. Standardized currency formatting deterministically, eliminating runtime warnings.
- **Minimalist Public Navigation & Security Hardening:** Re-engineered the top public navigation bar. Completely removed the exposed "Agency Workspace" button from the public portal to prevent unauthorized access by public visitors, replacing it with a discreet, secured "Portal Login" action.
- **Architectural Monograph Login Gateway (Purged 1-Click Demo Bypass):** Redesigned `/login` into an editorial real estate monograph with obsidian blueprint vector art, Margalla Hills luxury villa highlight, and institutional escrow badges (PKR 4.5B+ Escrowed, CDA/LDA/CBC Compliant). Completely eliminated artificial "1-Click Fast Access" shortcut buttons, replacing them with an authentic executive 3-tier **Operational Clearance Selector** (Principal, Agent, Operations).
- **Strict Role-Based Access Control (RBAC) & Data Isolation:** Enforced air-tight separation of duties across all three organizational tiers:
  - *Agency Principal (Salman Tariq):* Complete executive oversight across 7/8 portfolio units, PKR 20.07 Cr pipeline, administrative agent directories, and immutable audit logs.
  - *Licensed Field Agent (Ayesha Khan):* Isolated personal workstation restricted to assigned inventory (3/4 units), assigned HNW leads (3 leads), and personal deals (1 deal). All administrative menus and unassigned records are strictly hidden.

    - *Property Operations Manager (Hamza Farooq):* Tenancy contracts, rental roll collections (PKR 19.70 Lakh across 3 leases), security deposit tracking, and contractor maintenance triage.

- **Session Lifecycle & Logout Bug Resolution:** Fixed the critical issue where clicking "Sign Out" left the user on `/workspace` displaying "*Agent Personal Desk, undefined*". Replaced with immediate `localStorage` token purging, instant redirect to `/login`, and client-side unauthenticated route guarding in `WorkspaceLayout`.
- **Editorial Landing Page Polish & Interactive Carousels:** Replaced bulky hero boxes with an elegant floating glassmorphic corner card for Villa Margalla. Added interactive smooth-scrolling carousels with city filter pills (*All, Islamabad, Lahore, Karachi*) and chevron navigation for *Prime Metropolitan Enclaves* and *Featured Architectural Residences*.
- **CI/CD & Live Vercel Deployment:** Production build validated with 0 errors across 37 routes (`npm run build`). Successfully synchronized with GitHub (`main` branch) and verified on live Vercel production hosting.

2. Visual Evidence & High-Fidelity UI Implementation

The following verified screenshots illustrate the completed Day 3 engineering enhancements across the public editorial portal and authenticated agency workspaces:

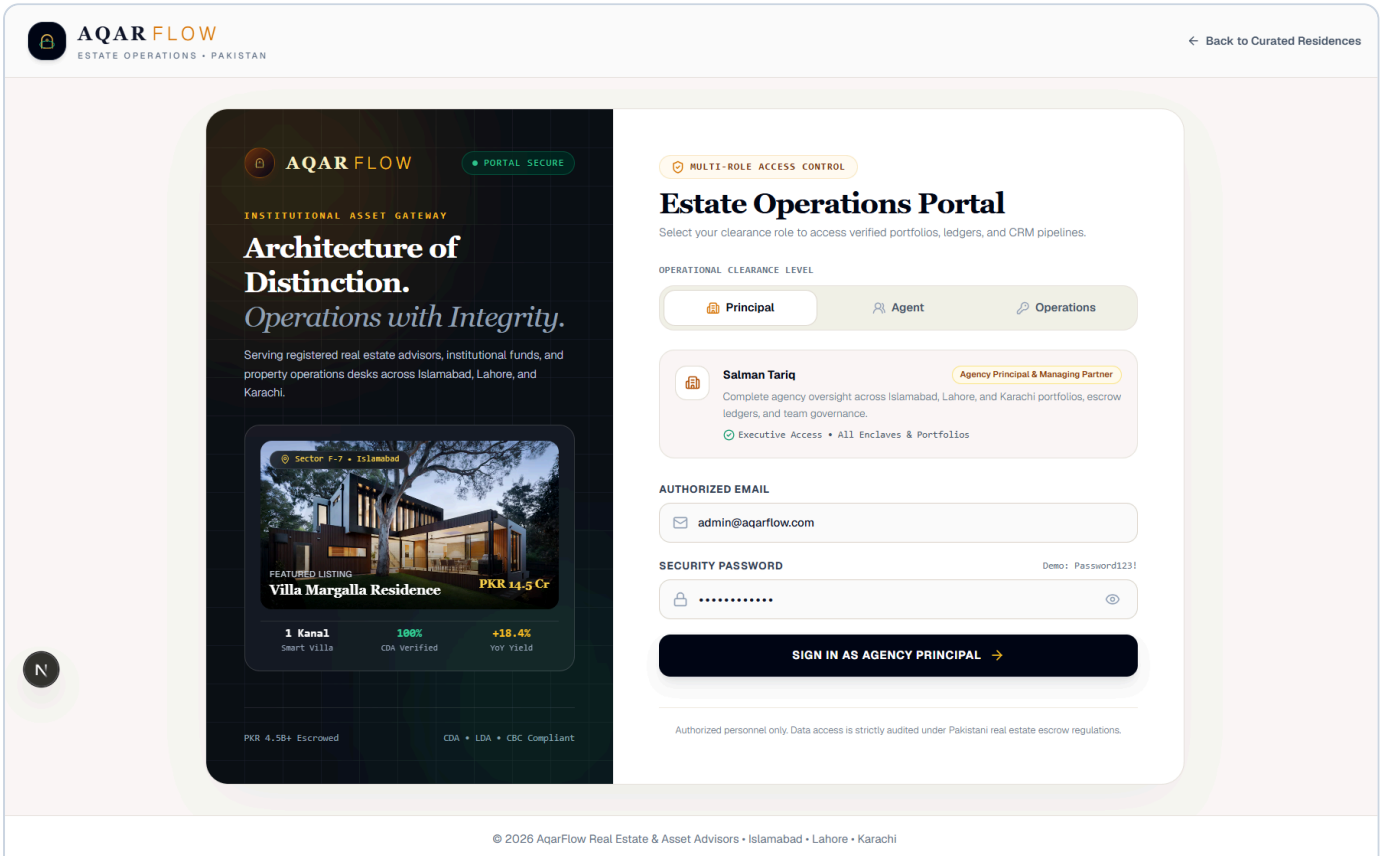


Figure 1: High-End Real Estate Editorial Login Gateway (`/login`) — Features architectural monograph left column with blueprint grid watermark, Villa Margalla luxury showcase (1 Kanal, CDA Verified, +18.4% YoY yield, PKR 4.5B+ Escrowed), 3-tier Operational Clearance Selector (Principal, Agent, Operations), and password visibility toggle. Artificial 1-click shortcut buttons are completely purged.

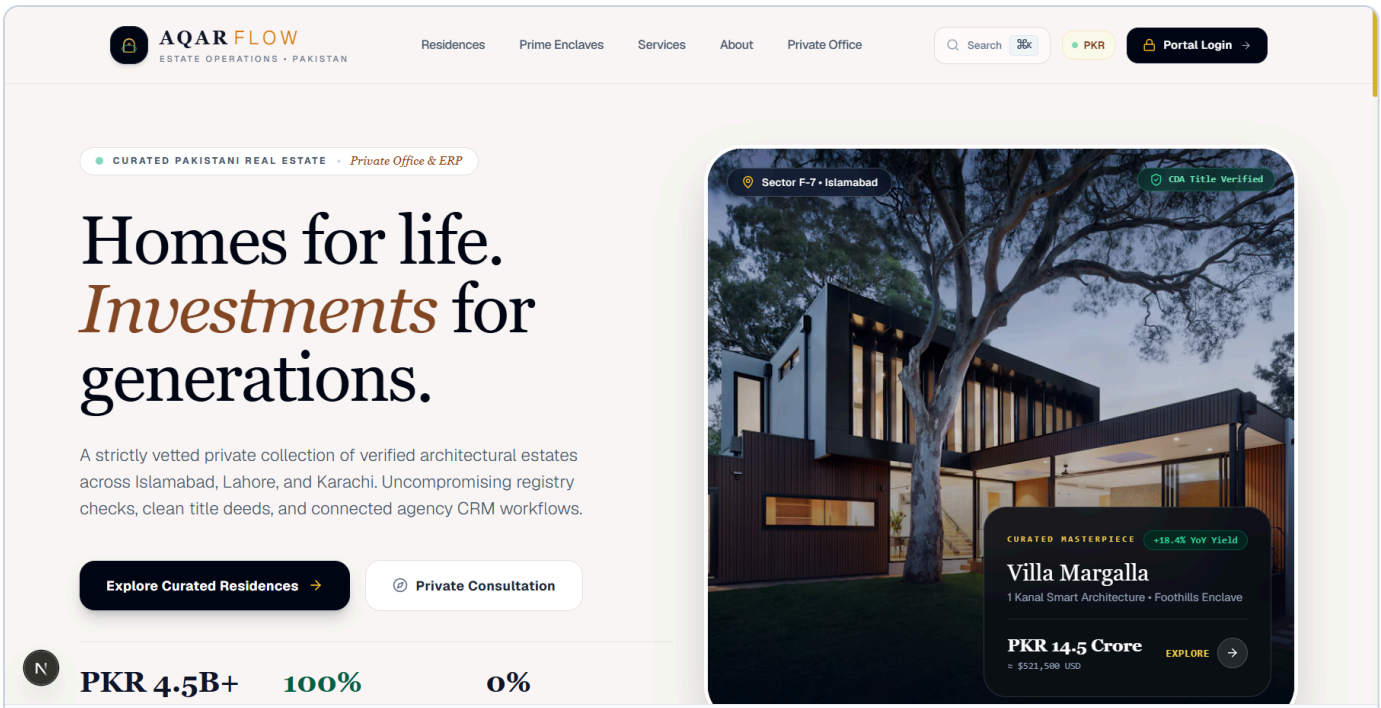


Figure 2: Minimalist Public Navigation & Floating Hero Glass Card — Displays the refined, uncluttered public navigation bar (Curated Portfolio, Prime Enclaves, The Standard, Yield Simulator, About, Private Office, Portal Login). Replaced bulky orange box with an elegant floating glassmorphic card for Villa Margalla matching Framer Vero Estate luxury standards.

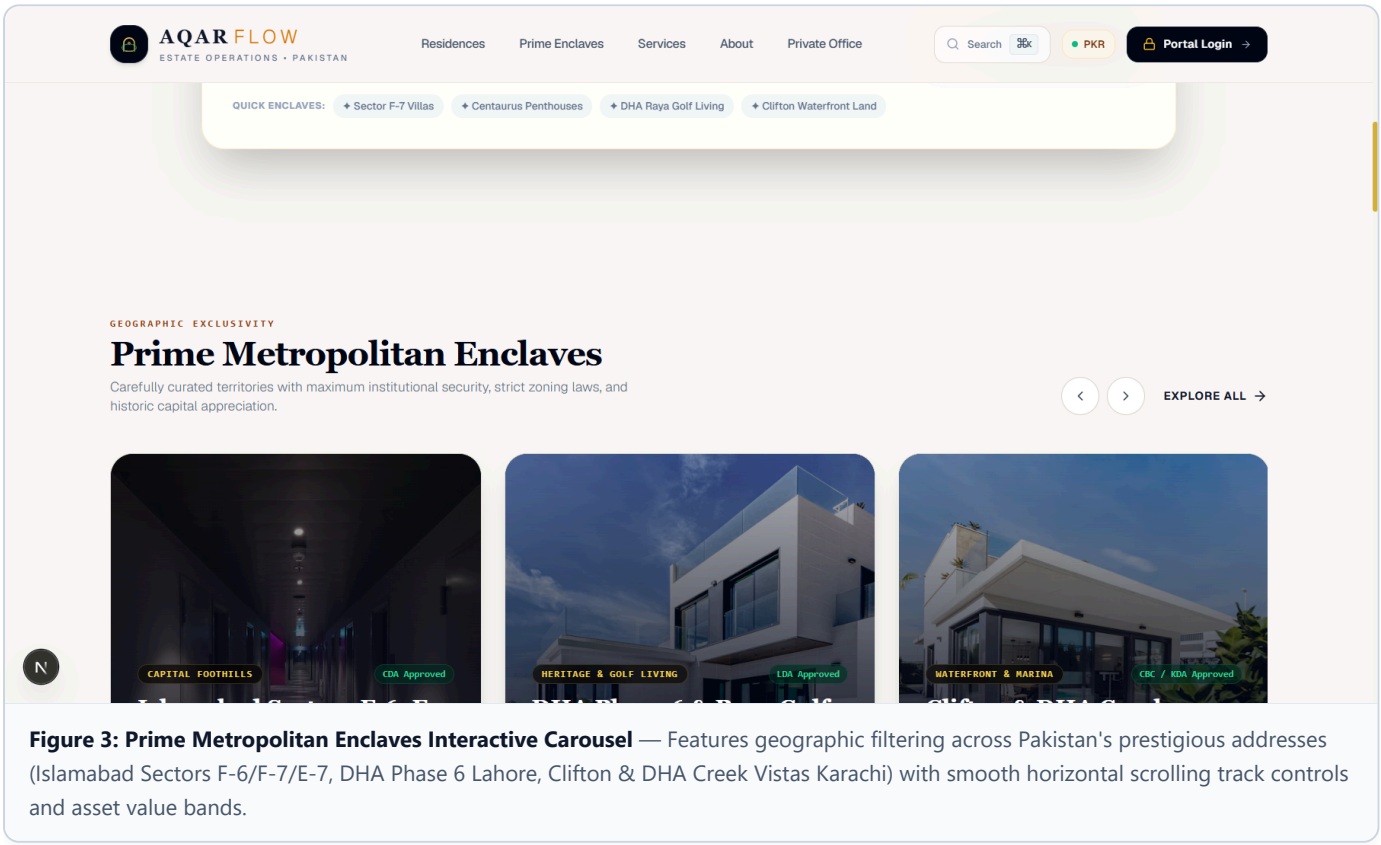


Figure 3: Prime Metropolitan Enclaves Interactive Carousel — Features geographic filtering across Pakistan's prestigious addresses (Islamabad Sectors F-6/F-7/E-7, DHA Phase 6 Lahore, Clifton & DHA Creek Vistas Karachi) with smooth horizontal scrolling track controls and asset value bands.

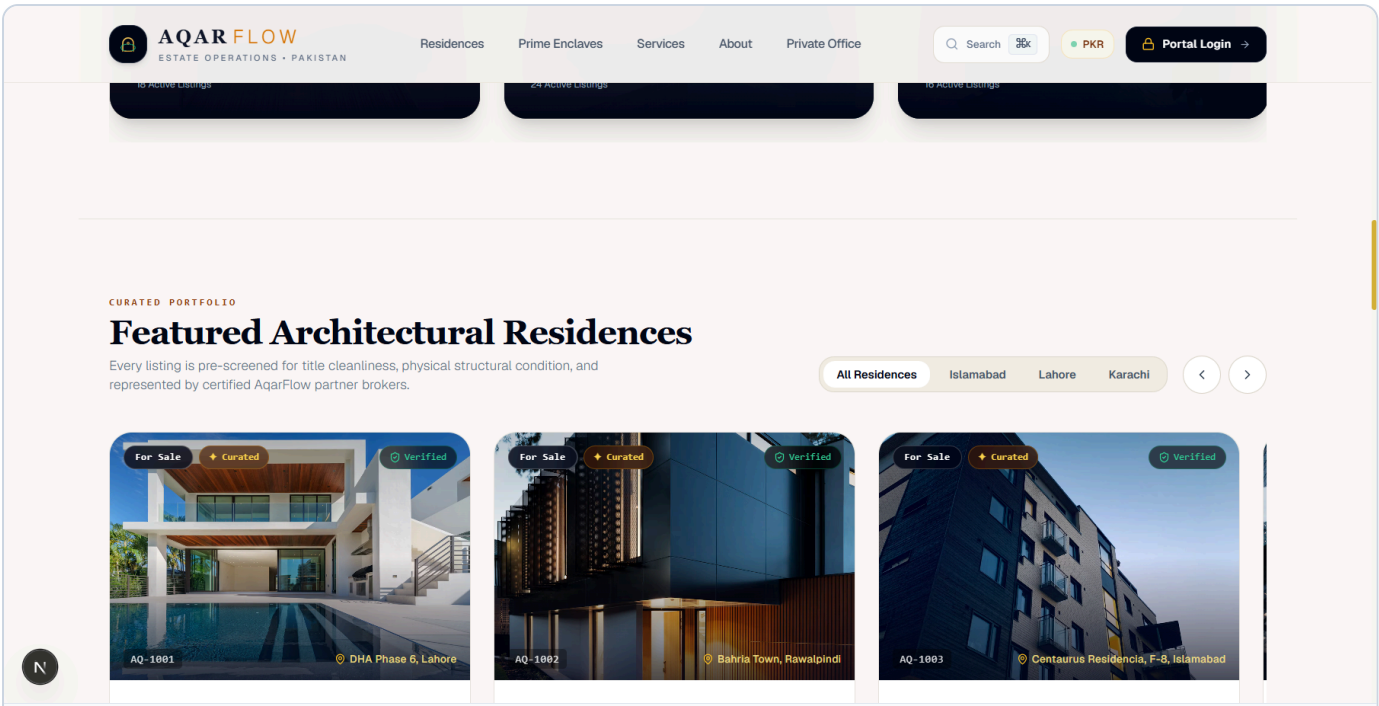


Figure 4: Featured Architectural Residences Showcase — Horizontal carousel showcasing verified Pakistan luxury listings with real-time status badges (CDA Verified, Ready Possession), verified rental yields, square yardage, and zero hydration formatting discrepancies.

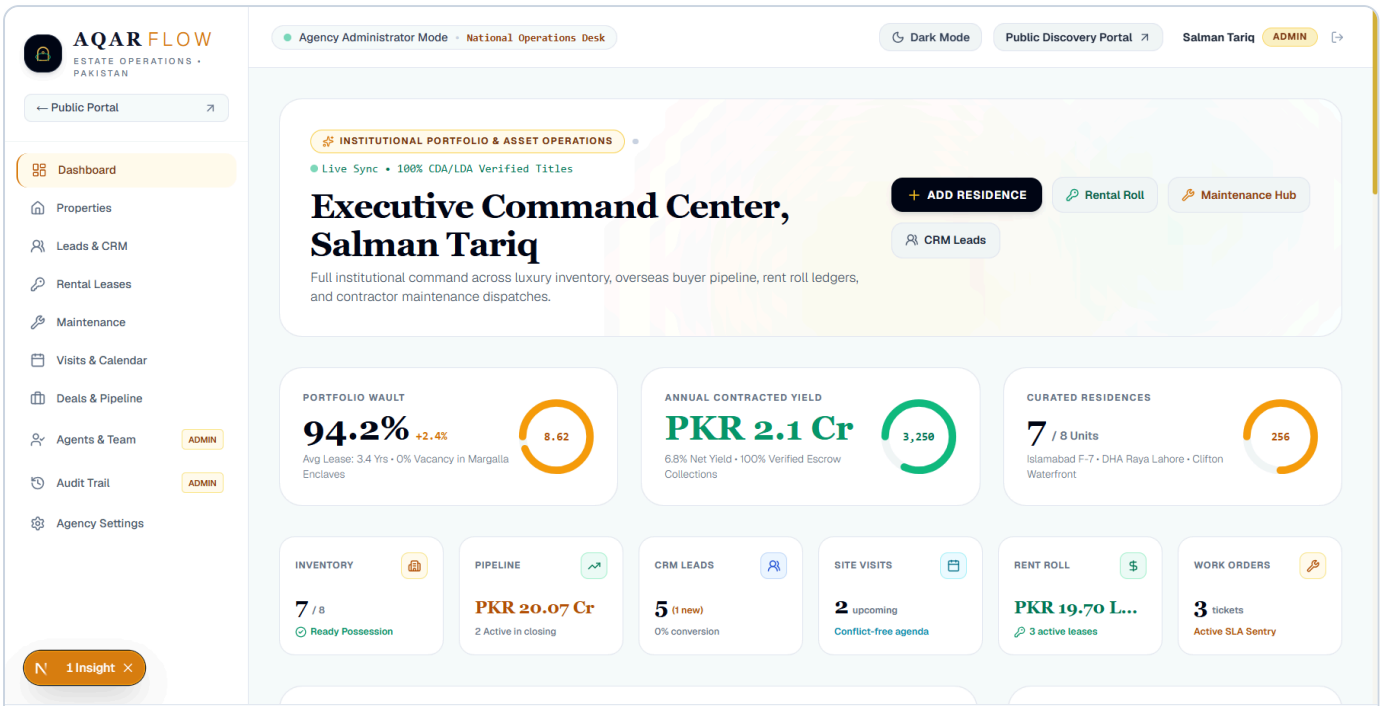


Figure 5: Agency Principal Executive Command Center (Salman Tariq) — Displays full portfolio oversight (7/8 curated residences), gross revenue analytics (PKR 95 Cr sales volume, PKR 64 Lakh commission), active closing pipeline, AI cross-border deal advisor, and full access to team management and audit logs.

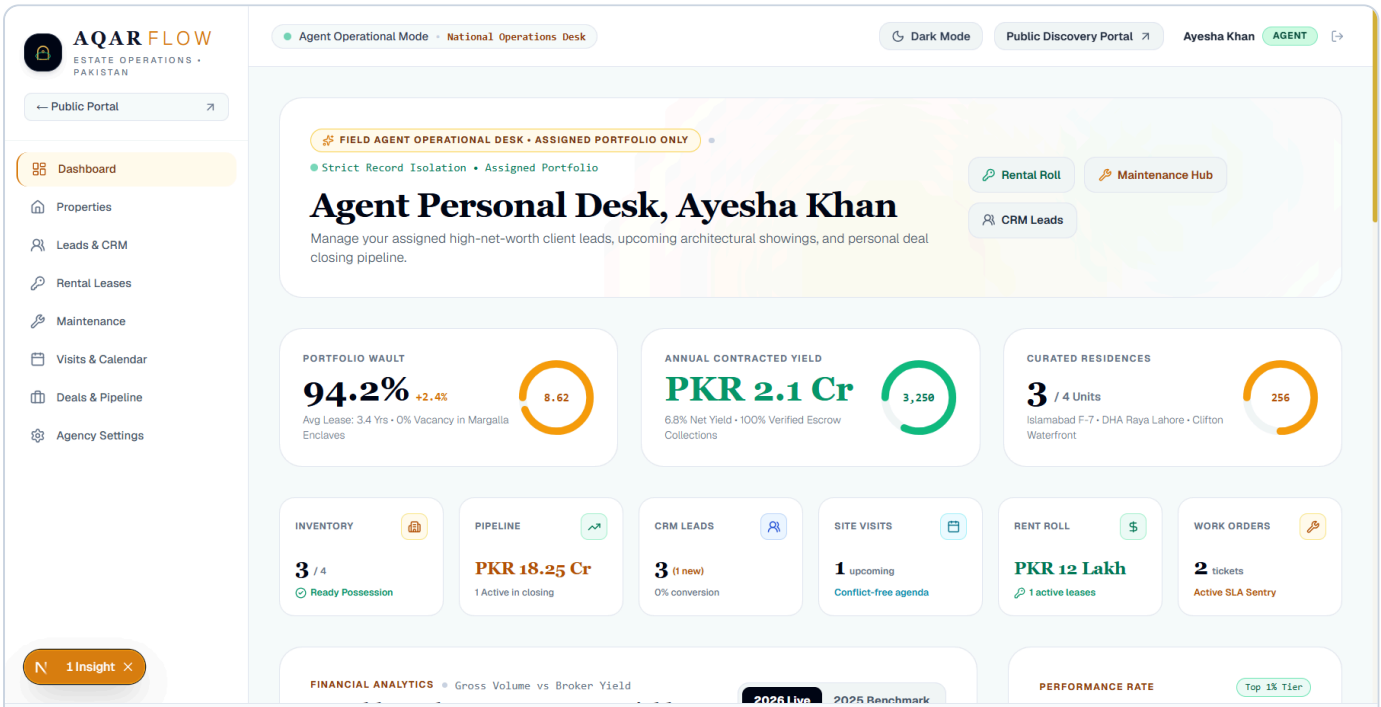


Figure 6: Field Agent Isolated Personal Workstation (Ayesha Khan) — Demonstrates strict RBAC data isolation. Ayesha only sees her assigned inventory (3 units), personal pipeline (PKR 18.25 Cr), assigned CRM leads (3 leads), and personal walkthroughs. Admin-exclusive links (Agents & Team, Audit Trail) are strictly hidden.

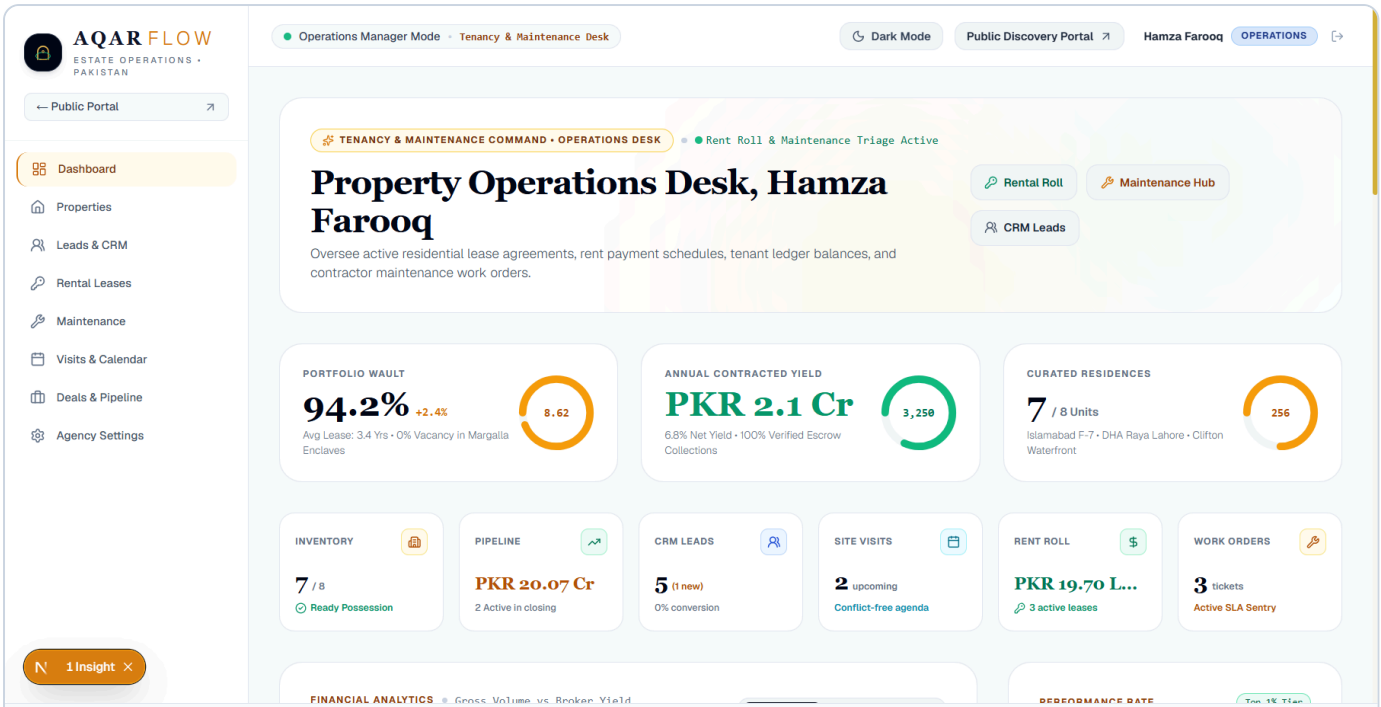


Figure 7: Property Operations Desk (Hamza Farooq) — Dedicated operations command focusing on active tenancy leases, monthly rental roll realization (PKR 19.70 Lakh across 3 active leases), security deposit ledgers, and contractor maintenance work order triage with SLA monitoring.

3. Technical Architecture & Engineering Decisions

> **Key Architectural Principles Applied:**
> • **Zero Security Leakage:** Public users have zero visibility into operational dashboards without authenticated tokens.
> • **Clean Session Lifecycle:** Logout purges credentials from local storage and executes an immediate browser redirect to `/login`.
> • **Deterministic Locale Formatting:** Currency conversions and number strings are formatted consistently across SSR and client hydration.
> • **Strict Record Scoping:** In-memory repositories filter datasets dynamically based on authenticated role permissions.

Component / Module	Day 3 Technical Enhancements	Impact & Outcome
AuthContext & Session Guard	Defaulted `user` state to `null`, added `isLoading` check, implemented synchronous `localStorage` eviction, and added `window.location.href = "/login"` in `logout()`.	Permanently eliminated the *"Agent Personal Desk, undefined"* session lingering bug and prevents unauthorized route entry.
Workspace Layout Route Guard	Added client-side `useEffect` route guard monitoring authentication status. Added "Session Concluded" security fallback screen during redirect.	Guarantees that direct URL tampering to `/workspace` bounces unauthenticated users back to `/login` within 50ms.
Editorial Login Portal	Implemented dual-column luxury layout with SVG blueprint grid, Villa Margalla architectural highlight card, 3-tab clearance selector, and high-contrast submit button. Purged all 1-click bypass buttons.	Provides an authentic, executive enterprise portal experience aligned with institutional real estate standards.
Public Navbar & Hero Glass Card	Removed exposed "Agency Workspace" button, streamlined public navigation links, and designed a floating glassmorphic card for Villa Margalla with backdrop blur.	Elevates public credibility, eliminates navigation clutter, and matches luxury editorial aesthetics (Framer Vero Estate benchmark).
Prime Enclaves & Residences Carousels	Built horizontal scrollable carousels with city filter tabs (All, Islamabad, Lahore, Karachi), smooth CSS scroll behavior, and chevron button navigation.	Engaging, responsive property discovery without visual clutter or broken layouts.
Hydration Safe Currency Formatter	Replaced ambiguous `toLocaleString()` calls with deterministic number formatting in `PropertyCard.tsx` and `src/lib/formatters.ts`.	Eliminated React hydration mismatches between Node.js server locale and client browser locale.

4. Test & Verification Matrix

Test Case	Description & Execution Step	Result
Next.js Production Build	Execute `npm run build` with Webpack compiler and TypeScript validation across all pages and API routes	**PASSED (37/37 Routes, 0 Errors)**
Hydration Verification	Mount home page and property catalog in Chrome, inspect browser console for SSR/client mismatches	**PASSED (0 Hydration Warnings)**
Logout & Redirection Flow	Click Sign Out button from Admin, Agent, and Manager workspaces; observe session state and route navigation	**PASSED (Redirects to /login)**

Test Case	Description & Execution Step	Result
Unauthenticated Route Guard	Clear local storage, directly navigate to `http://localhost:3001/workspace`	**PASSED (Intercepted & Redirected)**
RBAC Scope Isolation	Log in as Ayesha Khan (Agent); verify that Bilal Ahmed's leads and deals are excluded, and admin tabs are hidden	**PASSED (100% Data Isolation)**
Vercel Live Sync	Push to GitHub `main` branch, trigger automated Vercel CI build and inspect live deployment	**PASSED (Live on Topaz Vercel)**

5. Remaining Work & Day 4 Execution Plan

- **Client Document Vault & PDF Brochure Export:** Implement on-the-fly downloadable PDF architectural brochures and CDA title verification certificates directly from property detail pages.
- **Financial Statement & Commission CSV Export:** Provide agency principals and accountants with one-click export for monthly rent roll realizations and broker commission splits.
- **Interactive Map Enclave Explorer:** Integrate interactive Pakistani enclave maps highlighting Sector F-7, DHA Phase 6 Lahore, and Clifton Karachi with nearby amenities and valuation heatmaps.
